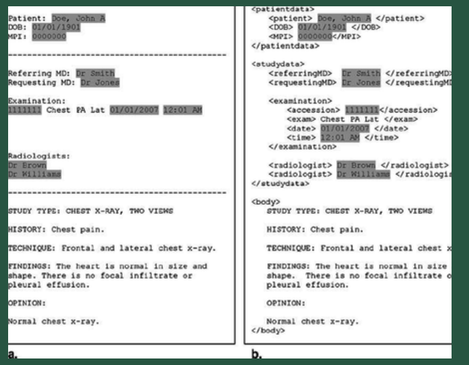
AI IN HEALTHCARE GROUP PROJECT



Using clinical reports free text to determine the level of difficulty for a computer vision model



Patient: Doe, John A
DOB: 07/07/1961
MPI: 0000000

Referring MD: Dr Smith
Requesting MD: Dr Jones

Examination:
111111 Chest PA Lat 01/01/2007 12:01 AM

Radiologists:
Dr Brown
Dr Williams

STUDY TYPE: CHEST X-RAY, TWO VIEWS

HISTORY: Chest pain.

TECHNIQUE: Frontal and lateral chest x-ray.

FINDINGS: The heart is normal in size and shape. There is no focal infiltrate or pleural effusion.

OPINION:

Normal chest x-ray.

a.

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<body>
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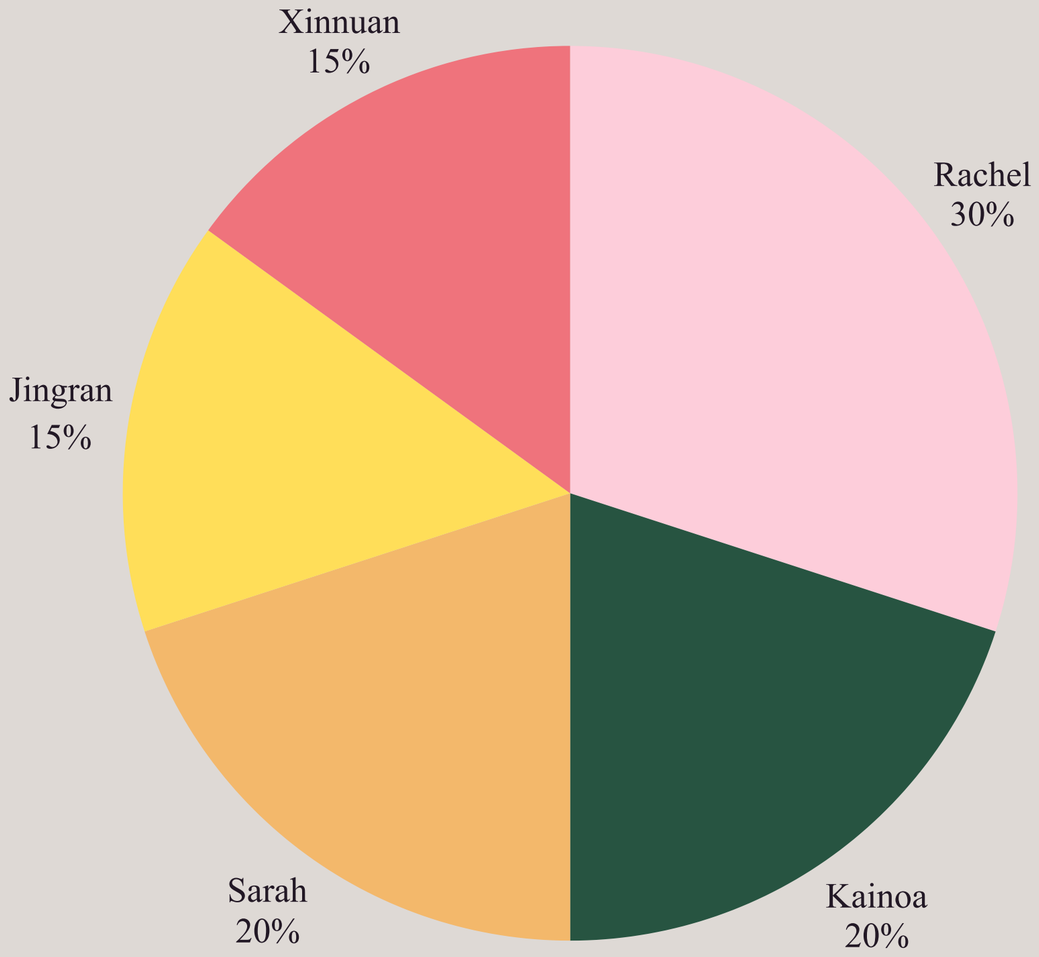
OPINION:

Normal chest x-ray.
</body>

b.

Rachel Lau
Kainoa Martinez
Sarah Lee
Jingran Zheng
Xinnuan Wang

WORK DISTRIBUTION



Rachel (30% / 12.0 points) — Led both CV and NLP models; edited all Python scripts for TRE; executed full pipeline; 13 presentation slides; co-created work distribution graphic.

Kainoa (20% / 8 points) — Project management & GitHub infrastructure; PCA coding; 4 presentation slides; slide graphics; contributed/consolidated literature review.

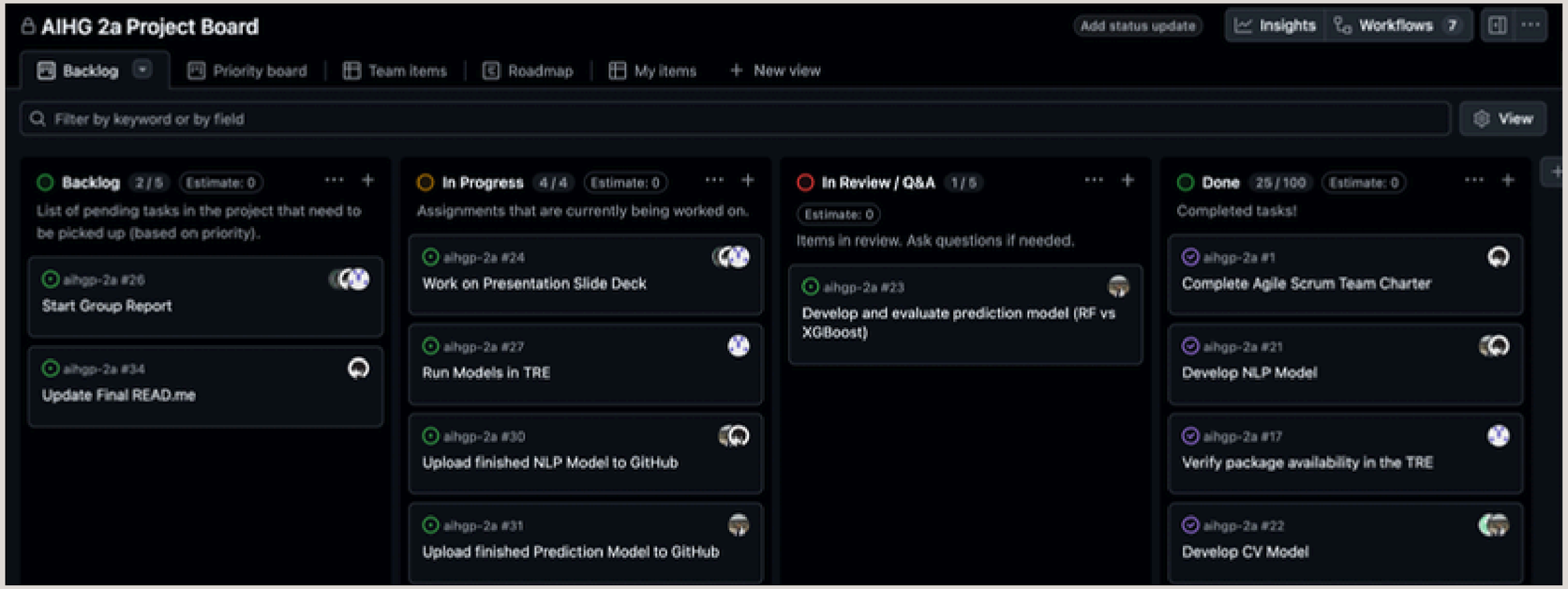
Sarah (20% / 8 points) — CV model development; 10 presentation slide; co-authored project management slides; contributed to consolidated literature review.

Jingran (15% / 6 points) — TRE data access and handling; running pipeline in TRE

Xinnuan (15% / 6 points) — Problem statement & literature review draft for group report.

PROJECT MANAGEMENT

- **Adopted the Kanban Agile approach (continuous flow)**
- **Technical Team & Literature Team**
  - **Task allocation based on members' strengths/comfortability**
- **Progress tracking via GitHub**
- **Weekly meetings – updates, discuss blockers and raise action points**



WORKFLOW

Background Research



Define Problem Statement



Model design and training



Deployment in TRE

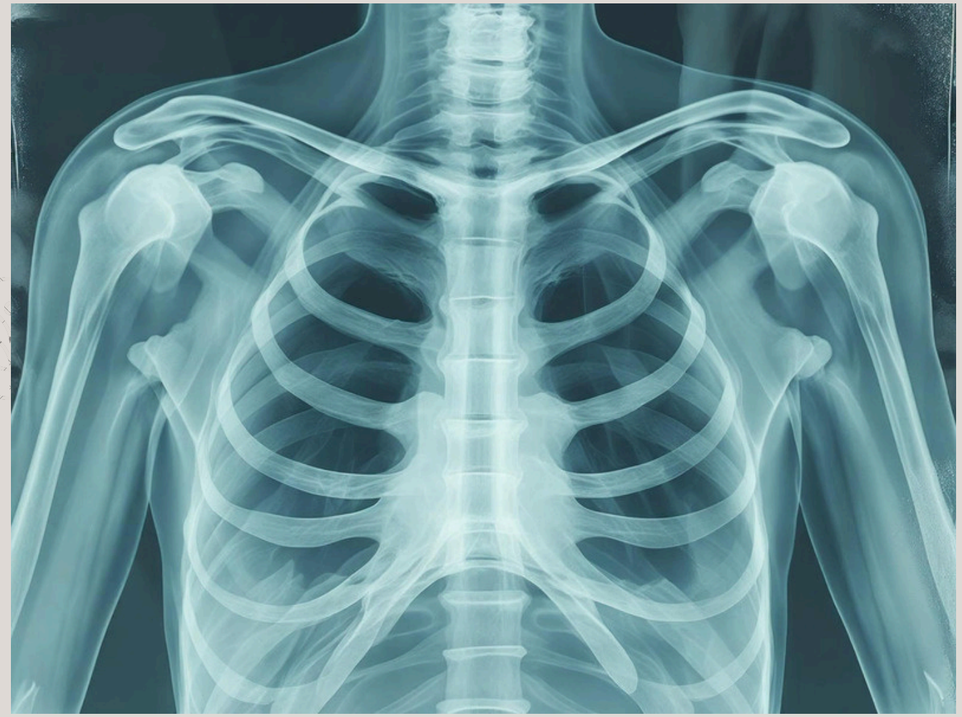


Analysis & Interpretation



BACKGROUND

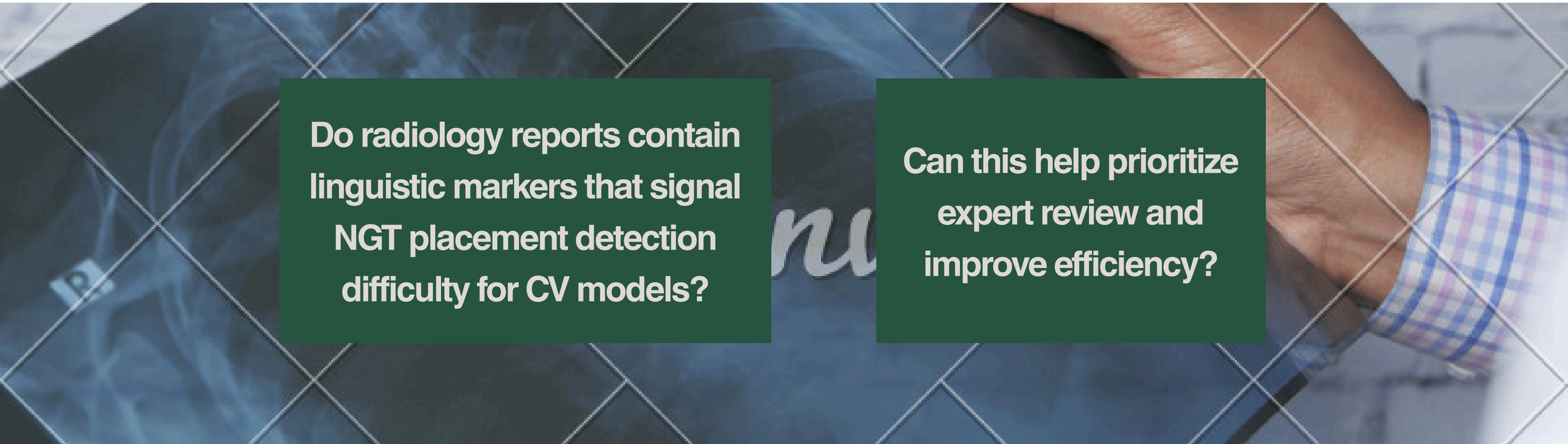
- **CV models are increasingly used for chest X-ray classification[1,3]**
- **Routine pairing of images & radiology reports[4,8]**
- **CV models produce confidence scores indicating classification certainty[3]**
- **Limited literature on predicting CV difficulty using clinical text[5,6]**
- **Clinical text may contain linguistic signals predictive of image classification difficulty[5,6]**
- **Nasogastric Tubes (NGT) placement verification is a critical safety task to ensure patient safety**



PROBLEM STATEMENT

CAN NATURAL LANGUAGE PROCESSING OF RADIOLOGY REPORTS PREDICT CHEST X-RAY IMAGE DIFFICULTY?

OUR INVESTIGATION



Do radiology reports contain linguistic markers that signal NGT placement detection difficulty for CV models?

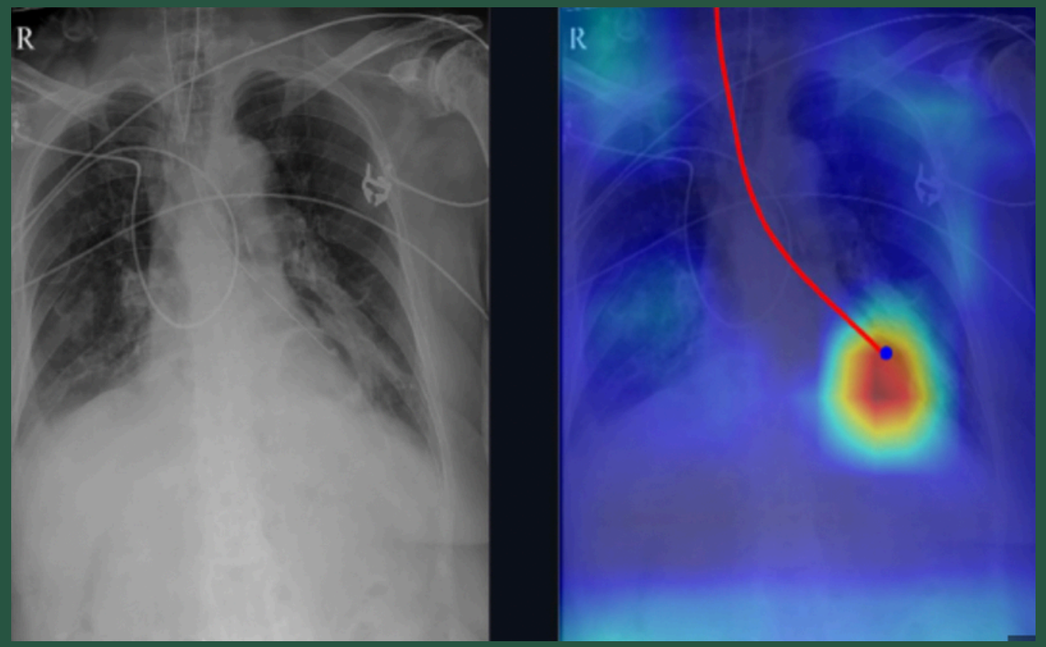
Can this help prioritize expert review and improve efficiency?

RATIONALE



- CV models show variable performance on chest X-rays
- Some X-ray images prove significantly harder to classify
- Safe clinical deployment requires understanding of image classification difficulty
- While difficulty is typically assessed visually, radiology reports may contain linguistic signals of difficult cases

PRACTICAL IMPLICATIONS



1. **Improve clinical workflow efficiency**

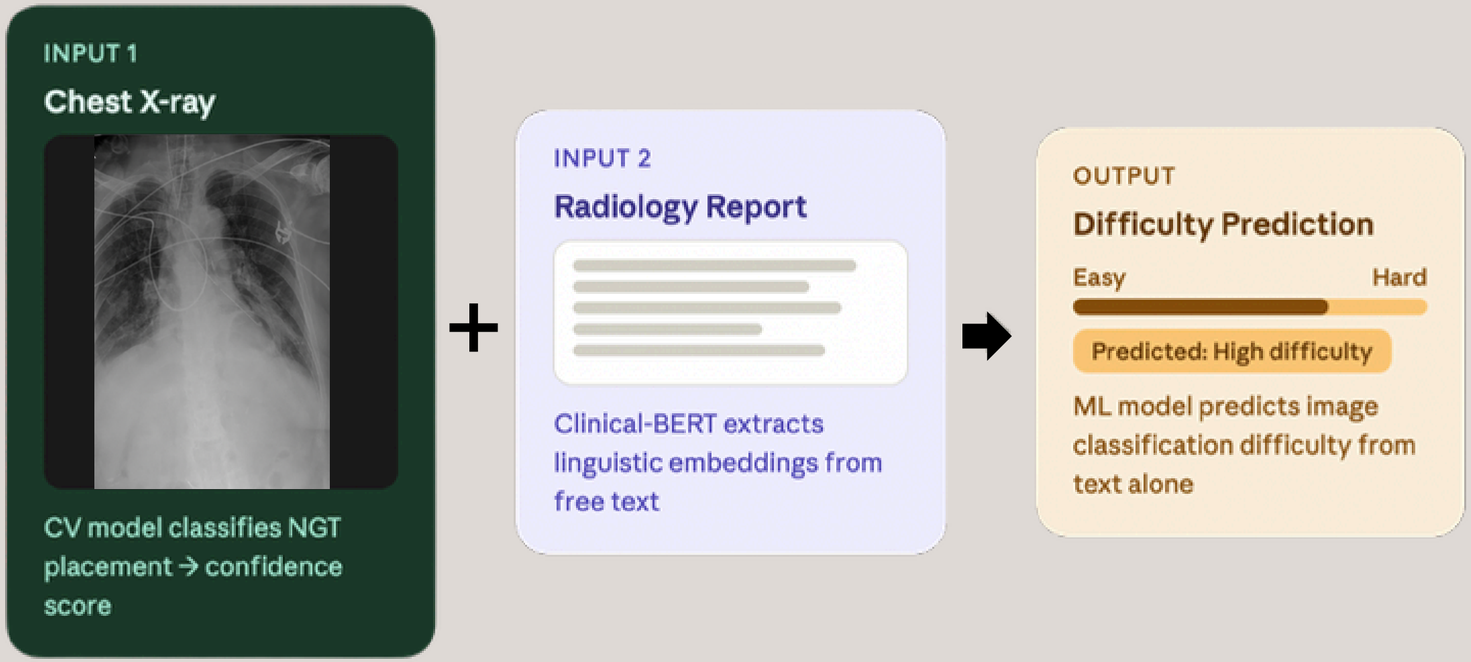
2. **Potential for generalizability to other medical imaging domains**

3. **Identification of difficult cases could prioritize human expert review**

PROJECT OUTLINE

Project Aim:

- **Develop a machine learning model that uses radiology report text to predict NGT placement detection difficulty for computer vision (CV) models.**



Research Question:

- Do radiology reports contain sufficient linguistic information to predict classification of NGT placement in chest X-rays for CV models?

DEFINITION OF DIFFICULTY

Difficulty = CV model's confidence in correctly identifying NGT placement status
(properly placed vs. misplaced)

EASY: High confidence (≥ 0.80)
→ Model is certain about its classification
HARD: Low confidence (< 0.80)
→ Model is uncertain, image may require expert review

PROJECT GOALS

Minimum Viable Product (MVP):

- A working binary classifier (EASY/HARD)
- Achieving at least 70% accuracy
- Validation on a proxy dataset before implementation in the TRE

Optimal Product (OP):

- A working multi-class classifier (EASY/MODERATE/HARD)
- Achieving at least 85% accuracy
- Validation on real clinical data within the TRE

PIPELINE OVERVIEW

(1) COMPUTER VISION MODEL

- **Model: Custom ResNet-50 binary classifier**
- Ran CV model on X-ray images to get predicted confidence scores

(2) NATURAL LANGAUGE PROCESSING (NLP)

- **Model: Clinical-BERT**
- Extracted semantic embeddings from radiology reports

(3) PREDICTION MODELLING

- **Model: Dual-Output Regression**
- **Input:** BERT embeddings
- **Output:** Confidence score
- Compared a linear regression and random forest model
- Trained to predict confidence scores from report text

METHODOLOGY - CV MODEL

Task:

- NGT placement detection (properly placed vs. misplaced)

Model:

- Custom ResNet-50 Binary Classifier

Output:

- Confidence scores (0–1) indicating model certainty in NGT placement predictions
- Used as ground truth proxy for image difficulty

Group 2a Presentation

METHODOLOGY - NLP

Model: ClinicalBERT

Pre-training:

- Trained on 2+ million clinical notes
- Captures domain-specific medical language patterns

Input:

- Free-text radiology reports paired with chest X-rays

Output:

- 768-dimensional contextual embeddings (one per report)
    - Each embedding encodes semantic meaning of the entire report

Rationale:

- Embeddings capture linguistic nuances predictive of image difficulty without requiring manual feature engineering
- Embeddings served as input features for prediction model

METHODOLOGY - PREDICTION MODEL

Model: Dual-Output Regression with Thresholding

- **Compared performance for linear regression and random forest model**

Architecture:

- Single linear layer: 768-dimensional input → continuous score output

Input:

- 768-dimensional BERT embeddings from radiology reports

Regression Output:

- Predicted confidence scores (0–1) for NGT placement detection
- Interpretation: Model learns which linguistic patterns correlate with CV model confidence

Classification:

- Difficulty labels derived from predicted confidence scores
- Thresholding: Predicted confidence ≥ 0.80 = EASY; < 0.80 = HARD

MODEL PERFORMANCE

```
Linear Regression:
  Regression Performance:
    MAE:  0.061 ± 0.003
    R²:   -2.325 ± 0.083
  HARD Class Performance:
    F1:       0.000 ± 0.000
    Recall:   0.000 ± 0.000
  Overall Performance:
    Balanced Acc: 1.000 ± 0.000
    MCC:          0.000 ± 0.000
```

```
Random Forest:
  Regression Performance:
    MAE:  0.033 ± 0.001
    R²:   -0.061 ± 0.034
  HARD Class Performance:
    F1:       0.000 ± 0.000
    Recall:   0.000 ± 0.000
  Overall Performance:
    Balanced Acc: 1.000 ± 0.000
    MCC:          0.000 ± 0.000
```

Our model's performance did not meet our MVP and optimal product accuracy goals

Regression Performance Metrics:

(1) Linear Regression:

- **MAE: 0.061**
- **R-squared: -2.325**

(2) Random Forest:

- **MAE: 0.033**
- **R-squared: -0.61**

Imbalanced Classes

- **100% of cases were classified as HARD**
- **Suggests that report text lacked predictive signals**

Both models revealed minimal prediction power of radiology report text for predicting CV model confidence scores

INTERPRETATION OF RESULTS

(1) Automated difficulty prediction may require multimodal approaches

- **Combine image quality metrics with text features**

(2) Radiology report text does not contain sufficient linguistic signals

- **Primarily describe clinical findings rather than image quality factors that drive model difficulty**

(3) Report text only approaches may be insufficient for CV performance prediction

LIMITATIONS

Binary classification (EASY/HARD)

- Lack of middle ground label
- Easier to misclassify label.

Ground Truth Dependency:

- Difficulty labels tied to one CV model's confidence scores
- Hence, may not generalize to other architectures

Clinical-BERT may not fully capture procedure-specific terminology common in chest X-ray NGT reports

LIMITATIONS

Data Subsampling:

- Processed subsample of TRE dataset due to NLP computational constraints
- May introduce selection bias
- May reduce statistical power for detecting linguistic patterns

Patient Matching Approach:

- TRE identifier systems incompatible
- Index-based matching used (both datasets seed=42)
- Potential impact: May introduce noise or obscure weak signals

CHALLENGES

(1) Model Pipeline Complexity

- Sequential dependencies: CV model → NLP → ML classifier
- Managing three models with different data requirements
- Ensuring compatibility across pipeline stages

(2) NLP Processing & Data Constraints

- Clinical-BERT embedding extraction computationally intensive
- Subsampling necessary to meet project timeline within TRE environment
- Balancing analytical depth vs. resource availability

(3) Implementation into the TRE

- Extracting and processing DICOM images from zipped archives
- Matching images to radiology reports

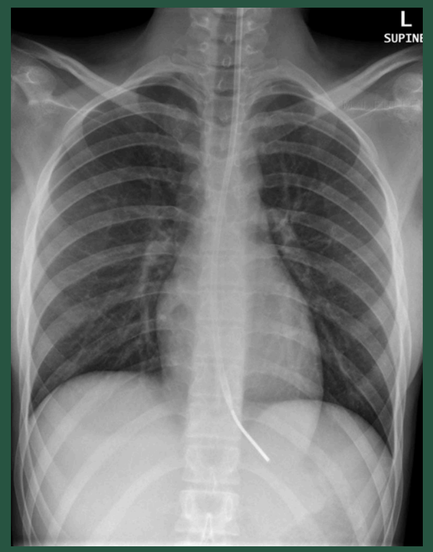
(4) Defining "Difficulty"

- Abstract concept → measurable metric
- Threshold selection: 0.80 cutoff vs. percentile-based splitting

FUTURE WORK

- Validate with NGT-specific model and patient-level ID matching
- Multi-class difficulty classification (EASY/MODERATE/HARD) with proper confidence calibration
- Develop multimodal prediction combining image quality metrics and clinical text features
- Fine-tune Clinical-BERT on domain-specific radiology language for domain adaptation
- Scale to full dataset (19K+ reports) with GPU optimization
- Extend validation to other imaging tasks and modalities

CONCLUSION



- Developed an end-to-end pipeline that predicts how difficulty of NGT placement detection from radiology report text alone
- A working binary class model was developed and validated on the TRE dataset

- MVP and OP accuracy goals not met
- Future work could focus on further validation and optimization of the model

Key Finding:

- Our results suggest that free text in radiology report does not contain meaningful linguistic signals

